

ACTIVITY OF INTEGRATION (S. 5)

PHYSICS: PROJECTILES

Scenario:

In a mountainous village with no mobile signal, a power surge has damaged the only radio communication device. A local rescue team suggests launching a signal flare toward a nearby military outpost on a plateau located 80 meters higher and 150 meters away horizontally.

The flare gun can launch at a speed of 45 m/s, and the rescue team wants your Physics class to determine:

- What launch angle is needed to reach the outpost?
- Will the flare go over or fall short of the target?
- Is there enough height in the trajectory?

This real-life crisis needs quick physics-based answers.

Tasks:

1. Assuming a launch angle of 40° , calculate:

i) the maximum height reached by the flare:

ii) If not, what minimum angle should be used to reach or exceed the outpost height?

2. Using the same 40° launch:

i) Find the horizontal range of the flare.

ii) Compute the total time of flight.